Markov Decision Processes

Our aim in this series of lectures is to extend our suite of heuristic search and optimization algorithms to the case where transitions to successor states are stochastic rather than deterministic: i.e., to problems characterized as sequential decision making *under uncertainty*. For example, we might be interested in planning an optimal route to work, or optimizing our strategy in the game of blackjack, or maximizing returns on a portfolio of investments in the stock market. All three of these examples involve stochasticity, as traffic, cards dealt, and stock prices are all unpredictable. This stochasticity can be estimated from data, as is common in the case of traffic or stock prices, or, e.g., in card games, deduced via combinatorics.

In these lectures, we introduce **Markov decision processes (MDPs)** as a modeling tool in the study of sequential decision making under uncertainty. They serve two primary purposes: predicting long-term rewards and computing optimal behavior, both in uncertain worlds. Solutions to MDPs may involve linear or dynamic programming, when the stochastic model is known and the state and decision spaces are sufficiently small; otherwise, they can be solved using reinforcement learning.

Our discussion of Markov processes is divided into two parts: the first part is concerned with computing state values V in Markov reward (or decision) processes, and the second with computing action values Q in Markov decision processes. This division coincides with two related problems, namely:

1. **(passive) prediction**, or **policy evaluation**: compute the state-value function V^π, given policy π
2. **(active) control**: find an optimal policy π^*, by computing the optimal action-value function Q^*

1 Markov Reward Processes

An agent operating in a stochastic environment transitions from state to state, in general obtaining rewards along the way, as follows: at time t,

1. state is s_t
2. receive reward r_t
3. transition to state s_{t+1} with probability $P[s_{t+1} \mid s_t, \ldots, s_0]$

We model this agent's interactions as a (discrete-time) **Markov reward process**, a tuple $\langle S, R, P \rangle$, where time is discrete: i.e., $t \in T = \{0, 1, \ldots\}$, and

- S is a finite set of states
- $R : S \to \mathbb{R}$ is a reward function
- $P : S \to \Delta(S)$ is a probability transition function

[1]These notes were compiled in conjunction with Professor Eric Ewing.

N.B. The notation $\Delta(X)$ denotes the set of probability distributions over a finite set X.

The probability function P describes transitions from one state to another. In general, P may depend on the past history of states. For example, the probability that a machine in a factory fails may depend on how long the machine has been in service. Likewise, demand for a product may depend on long-term trends, such as the season, fashion trends, or brand loyalty.

The key assumption defining a **Markov process** is that the probability transition function P is a function of only the current state, not the past history of states. In the factory example, the current state might be only whether the machine is currently operating or whether it has failed; in the product demand example, it may be only current sales and inventory.

N.B. Markov reward processes can have stochastic rewards as well as stochastic transitions. Our framework is sufficiently general, because MRPs with stochastic rewards can be reduced to MRPs processes with deterministic rewards by setting the deterministic rewards equal to the expected stochastic rewards.

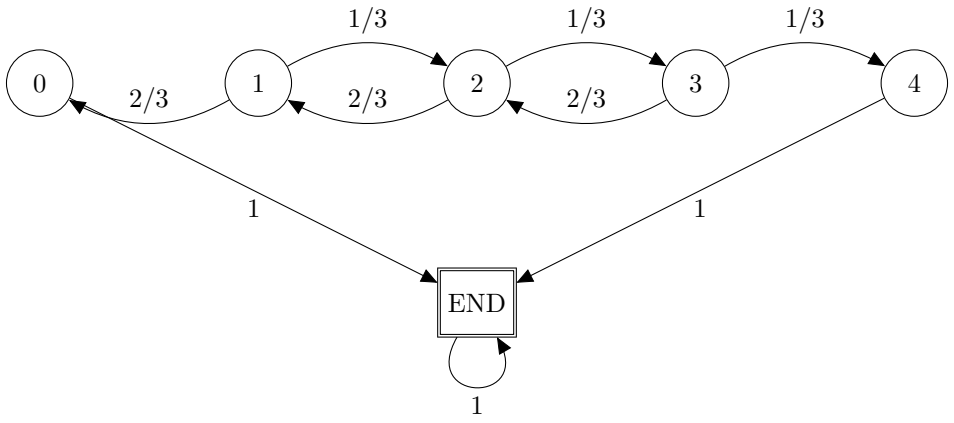


Figure 1: Gambler's Ruin: $N = 4$.

Example *Gambler's Ruin* is an example of a Markov reward process. A gambler gambles until he either wins a set amount of money, say $\$N$, or loses all his money. At state s_t, his wealth increases by \$1 with probability 1/3, and it decreases by \$1 with probability 2/3.

The set of states is defined by the worth of the gambler: $S = \{0, \ldots, N, \text{END}\}$. The transition probabilities are s.t. $P[i+1|i] = 1/3$ and $P[i-1|i] = 2/3$, for $i = 1, \ldots, N-1$, $P[\text{END}|i] = 0$, for $i = 0, N$, and $P[\text{END} \mid \text{END}] = 1$. What is the probability that the gambler wins, i.e., reaches state N?

Like all Markov chains, the Gambler's Ruin can be depicted as a graph, as in Figure 1, or as a Markov matrix. In this matrix, entries indicate the probability of transitioning from column j to row i:

$$
\begin{array}{r} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ \text{END} \end{array}
\begin{array}{c}
\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & \text{END} \end{array} \\
\left[\begin{array}{cccccc}
0 & \frac{2}{3} & 0 & 0 & 0 & 0 \\
1 & 0 & \frac{2}{3} & 0 & 0 & 0 \\
0 & \frac{1}{3} & 0 & \frac{2}{3} & 0 & 0 \\
0 & 0 & \frac{1}{3} & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{3} & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 1
\end{array}\right]
\end{array}
$$

Remark An **absorbing state** s is one where $P[s \mid s] = 1$. END is an absorbing state in the Gambler's Ruin.

2 State Values

Being a stochastic process, there are myriad trajectories that can be realized in a Markov reward process, each one with associated probabilities and rewards. The expected value over the rewards of all trajectories originating at a state s defines the **state-value function** $V : \mathcal{S} \to \mathbb{R}$. Given a Markov reward process, such as the Gambler's ruin, we are interested in computing V.

We write S_t to represent the random variable[2] denoting the state at time t: e.g., $S_t = s$, for some $s \in \mathcal{S}$. We then write $\tau_t = (S_t, S_{t+1}, \ldots)$ to denote the random trajectory originating at state S_t at time t, and $G_t^\tau = \sum_{i=0}^{\infty} R_{t+i}^\tau$ to denote the return accrued along trajectory τ: i.e., the sum of all the rewards from time t on, where $R_{t+i}^\tau = R(S_{t+i})$, for all $i \in \mathbb{N}$. The expected value of G_t^τ is then $V(s)$, where the expectation is taken over all trajectories τ that initiate at s: i.e., $V(s) = \mathbb{E}_\tau \left[G_t^\tau \mid S_t = s \right]$.

In this section, we present a proof sketch of Bellman's seminal theorem, known as **Bellman's equation**, namely the state value $V(s_t)$ can be decomposed recursively into the sum of the immediate reward obtained at time t and the discounted sum of the expected future rewards obtained thereafter, i.e., $V(S_{t+1})$.

2.1 But First . . . A Word on Return

Given trajectory $\tau = (S_t, S_{t+1}, S_{t+2}, \ldots)$, the return G_t^τ is a function of the current reward R_t and the stream of future rewards $R_{t+1}, R_{t+2}, \ldots$. In the case of a finite horizon, say of length $T < \infty$, return can be computed simply as the sum of current and future rewards: i.e.,

$$G_t^\tau = R_t + R_{t+1} + R_{t+2} + \ldots + R_T = \sum_{i=0}^{T-t} R_{t+i} \tag{1}$$

In the case of infinite horizons, however, the sum of future rewards is potentially infinite. If all trajectories are **proper** (as in the Gambler's ruin; a trajectory is called **proper** iff it eventually transitions to a zero-reward absorbing state), return can be computed simply as the sum of current and future rewards, as in Equation 1. Otherwise, return is computed as the sum of current rewards and the **discounted** sum of future rewards: i.e., assuming discount factor $0 \leq \gamma < 1$,

$$G_t^\tau = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \ldots = \sum_{i=0}^{\infty} \gamma^i R_{t+i} \tag{2}$$

If rewards are assumed to be bounded, return as defined by Equation 2 is finite. (**Exercise**) But even in the case of finite horizons or proper trajectories, G_t^τ is often computed with discounting, because of the following economic intuition.

The motivation for discounting future rewards can be simply stated: *a dollar today is worth more than a dollar tomorrow.* In particular, given an **interest rate** of $\rho\%$ per annum, the D dollars today are worth $(1 + \rho)D$ dollars a year from now. Likewise, D dollars a year from now are worth only $D/1+\rho = \gamma D$ dollars today, where γ denotes the **discount factor**, which is inversely related to the interest rate as $\gamma = 1/1+\rho$.

In other words, the **future value** of D dollars today is $(1 + \rho)D$, while the **present value** of D dollars tomorrow is γD. More generally, the future value of D dollars n periods into the future is $(1 + \rho)^n D$, while the **present value** of D dollars n periods into the future is $\gamma^n D$.

For example, with an interest rate of $\rho = 0.1$, the discount factor $\gamma \sim 0.909$, so that \$100 next year is worth $\gamma D \sim$ \$91 today and \$100 in two years is worth $\gamma^2 D \sim$ \$82 today.

[2] As usual, we use capital letters to denote random variables, and lower case letters to denote their realized values.

When the interest rate is high, say $\rho = 1$ (i.e., 100% interest rate) and $\gamma = 1/2$, \$100 today will be worth \$200 tomorrow; in this case, it is worth grabbing the \$100 immediately, so that it can accrue interest. On the other hand, when the interest rate is low, say $\rho = 0$ and $\gamma = 1$, present value equals future value; in this case, nothing is gained by collecting the rewards today rather than waiting until tomorrow.

Intuitively, ρ, and therefore γ, determine the relative worth of immediate vs. future rewards. As $\gamma \to 0$, immediate rewards are deemed more and more relevant; agents that attempt to maximize return in these circumstances are called **myopic**. As $\gamma \to 1$, future rewards are weighted more and more heavily; agents that aim to maximize discounted rewards based on high values of γ exhibit **foresight**.

Remark Assuming a finite interest rate $\rho \geq 0$, $\gamma \in (0, 1]$.

2.2 Bellman's Theorem

We now derive Bellman's theorem for Markov reward processes: the state value $V(s_t)$ at state s_t is the sum of the immediate reward obtained at time t and the discounted sum of the expected rewards obtained thereafter: i.e., for $0 \leq \gamma < 1$,

$$V(s_t) = r_t + \gamma \mathbb{E}_{s_{t+1}}[V(s_{t+1})] \tag{3}$$

The crux of the proof of this theorem is roughly as follows: for all states $s \in \mathcal{S}$,

$$\begin{aligned}
V(s) &= \mathbb{E}_\tau \left[G_t^\tau \mid S_t = s \right] \\
&= \int_\tau P[\tau \mid S_t = s] G_t^\tau \\
&= r_t + \gamma \int_\tau P[\tau \mid S_t = s] G_{t+1}^\tau \\
&= r_t + \gamma \sum_{s' \in \mathcal{S}} P[S_{t+1} = s' \mid S_t = s] \left(\int_{\tau'} P[\tau' \mid S_{t+1} = s', S_t = s] G_{t+1}^{\tau'} \right) \\
&= r_t + \gamma \sum_{s' \in \mathcal{S}} P[S_{t+1} = s' \mid S_t = s] \left(\int_{\tau'} P[\tau' \mid S_{t+1} = s'] G_{t+1}^{\tau'} \right) \\
&= r_t + \gamma \sum_{s' \in \mathcal{S}} P[S_{t+1} = s' \mid S_t = s] \mathbb{E}_{\tau'} \left[G_{t+1}^{\tau'} \mid S_{t+1} = s' \right] \\
&= r_t + \gamma \sum_{s' \in \mathcal{S}} P[S_{t+1} = s' \mid S_t = s] V(s') \\
&= r_t + \gamma \mathbb{E}_{s'}[V(s')]
\end{aligned}$$

The first equality follows from the definition of V; the second, from the meaning of an expectation; the third, from the fact that r_t is not a random quantity; the fourth, from the definition of joint probability; the fifth, from the Markov property; the sixth, from the meaning of an expectation (again); the seventh, from the definition of V (again); and the last line is simply an abbreviation.

2.3 Bellman's Equations

Bellman's theorem gives rise to the following system of $|S|$ equations with $|S|$ unknowns, known as Bellman's equations: for all states $s \in \mathcal{S}$,

$$V(s) = r(s) + \gamma \sum_{s'} P[s' \mid s] V(s') \tag{4}$$

Assuming a finite state state, with n states, we can rewrite this equation in matrix notation as follows: for value and reward vectors $v, r \in \mathbb{R}^n$ and $n \times n$ transition probability matrix P,

$$v = r + \gamma P v \tag{5}$$
$$v - \gamma P v = r \tag{6}$$
$$Iv - \gamma P v = r \tag{7}$$
$$(I - \gamma P)v = r \tag{8}$$
$$v = (I - \gamma P)^{-1} r \tag{9}$$

Therefore, Bellman's equation can be solved by matrix inversion. But just as in linear regression, where we tend to prefer running gradient descent to computing a closed-form solution directly, iterative methods are the norm for solving Bellman's equations. Iterative methods are preferable when the system of equations is large—making matrix inversion expensive not only in terms of time, but even more so in terms of space—and because iterative methods are more robust to numerical approximation errors.

3 Policy Evaluation

Policy evaluation is a dynamic programming method that computes state values via iterative updates based on Bellman's equations:

$$V(s) \leftarrow r(s) + \gamma \sum_{s'} P[s' \mid s] V(s') \tag{10}$$

This update rule is described in pseudocode in Table 1.

POLICY_EVALUATION(MRP, γ, ϵ)	
Inputs	discount factor γ
	convergence test ϵ
Output	state-value function V
Initialize	$V = 0$ and $V' = \infty$

while $\max_s |V(s) - V'(s)| > \epsilon$ **do**

1. $V' = V$
2. for all $s \in \mathcal{S}$
   (a) $V(s) = r(s) + \gamma \sum_{s'} P[s' \mid s] V'(s')$

return V

Table 1: Policy Evaluation.

Gauss-Seidel's version of this algorithm incorporates in-place updating: i.e., updating with V, as shown in Table 2, rather than V', as shown in Table 1.

Let's run through these algorithms on the Gambler's Ruin, assuming rewards 0 everywhere except at state N, where the reward is 1.

Assuming $\gamma = 1$, these values can be computed via policy evaluation as follows:

GAUSS_SEIDEL(MRP, γ, ϵ)

Inputs	discount factor γ
	convergence test ϵ
Output	state-value function V
Initialize	$V = 0$ and $V' = \infty$

`while` $\max_s |V(s) - V'(s)| > \epsilon$ `do`

1. $V' = V$
2. for all $s \in \mathcal{S}$
   (a) $V(s) = r(s) + \gamma \sum_{s'} P[s' \mid s] V(s')$

`return` V

Table 2: Gauss-Seidel.

V	0	1	2	3	4	END
0	0	0	0	0	0	0
1	0	0	0	0	1	0
2	0	0	0	$\frac{1}{3}$	1	0
3	0	0	$\frac{1}{9}$	$\frac{1}{3}$	1	0
4	0	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{11}{27}$	1	0
5	0	$\frac{1}{27}$	$\frac{13}{81}$	$\frac{11}{27}$	1	0

With in-place computation *á la* Gauss-Seidel, working backwards from END to state 4 down to state 0, the computation proceeds as follows:

V	0	1	2	3	4	END
0	0	0	0	0	0	0
1	0	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	0
2	0	$\frac{13}{243}$	$\frac{13}{81}$	$\frac{11}{27}$	1	0
3	0	$\frac{133}{2187}$	$\frac{133}{729}$	$\frac{107}{243}$	1	0
100	0	0.0667	0.2	0.4667	1	0

Since rewards are 0 everywhere unless the gambler is *not* ruined, in which case rewards are 1, the final state values can be interpreted as the probability that the gambler is *not* ruined. The value function reveals that the gambler is more likely to be ruined than not at all states $1, \ldots, N-1$.

A Markov Chains

A **stochastic process** is a sequence of random variables $\{X_t\}_{t=0}^{\infty}$. A stochastic process $\{X_t\}_{t=0}^{\infty}$ induces a probability transition function $P[X_{t+1} = s_{t+1} \mid X_t = s_t, \ldots, X_0 = s_0]$: i.e., the probability that the state at future time $t+1$ is s_{t+1}, given that the states at past times $t, \ldots, 0$ were $s_t, \ldots, s_0$, respectively.

A **Markov process** is a stochastic process *s.t.* for all t, for all $s_0, \ldots, s_t, s_{t+1}$,

$$P[X_{t+1} = s_{t+1} \mid X_t = s_t, \ldots, X_0 = s_0] = P[X_{t+1} = s_{t+1} \mid X_t = s_t] \tag{11}$$

Equation 11 is the **Markov property**, sometimes called the memoryless property. It states that the future is independent of the past, given the present. Mathematically, probability transitions to future states, such as s_{t+1}, are conditionally independent of the past, $s_{t-1}, \ldots, s_0$, given the present state s_t.

A **time-homogeneous** Markov process is one where the probability of transitioning from one state to another in n steps is always the same, regardless of absolute time: for all times t, t' and states s, s',

$$P[X_{t+n} = s' \mid X_t = s] = P[X_{t'+n} = s' \mid X_{t'} = s] \tag{12}$$

A time-homogeneous Markov process over a discrete state space is called a **Markov chain**. Markov chains can be represented as row- (or column-)stochastic probability transition matrices.

B Convergence of Policy Evaluation

In this appendix, we show that Bellman's system of equations (Equation 4) has a solution, and that policy evaluation converges to this solution.

To solve Bellman's equations, we rely on Banach's fixed point theorem, also called the contraction mapping theorem. Given a metric space[3] (X, d), a mapping $f : X \to X$ is called a **contraction** iff there exists some $0 \leq k < 1$ *s.t.* $d(f(x), f(y)) \leq kd(x, y)$, for all $x, y \in X$.

Theorem [Banach] Given a complete[4] metric space (X, d) and a contraction mapping $f : X \to X$,

(i) there exists a unique $x^* \in X$ *s.t.* $f(x^*) = x^*$

(ii) for arbitrary $x^0 \in X$, the sequence $\{x^n\}$ defined by $x^{n+1} = f(x^n) = f^{n+1}(x^0)$ converges to x^*

To solve Bellman's equations, we define the mapping $f : \mathbb{R}^S \to \mathbb{R}^S$ as follows:

$$(f(x))(s) = r(s) + \gamma \sum_{s'} P[s' \mid s]x(s') \tag{13}$$

Theorem The mapping f is a contraction on $(\mathbb{R}^S, L_\infty)$.

[3] A metric space (X, d) is a set X together with a distance function $d : X \times X \to \mathbb{R}$ that satisfies: (i) $d(x, x) = 0$, for all $x \in X$; (ii) $d(x, y) = d(y, x)$ for all $x, y \in X$; and (iii) the triangle inequality—$d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$.

[4] An example of a complete metric space is $\mathbb{R}$.

Proof Let the metric d be the L_∞, or max, norm: i.e., $||x - y|| = \max_i |x_i - y_i|$. For all $x, y \in X$, and for arbitrary state $s \in \mathcal{S}$,

$$\begin{aligned}
& |(f(x))(s) - (f(y))(s)| \\
= & \left| r(s) + \gamma \sum_{s'} P[s' \mid s] x(s') - \left(r(s) + \gamma \sum_{s'} P[s' \mid s] y(s') \right) \right| \\
= & \gamma \sum_{s'} P[s' \mid s] \, |x(s') - y(s')| \\
\leq & \gamma \sum_{s'} P[s' \mid s] \max_{s''} |x(s'') - y(s'')| \\
= & \gamma \sum_{s'} P[s' \mid s] ||x - y|| \\
= & \gamma ||x - y||
\end{aligned}$$

It follows that $|(f(x))(s) - (f(y))(s)| \leq \gamma ||x - y||$, for all states s. Therefore, $||f(x) - f(y)|| = \max_s |(f(x))(s) - (f(y))(s)| \leq \gamma ||x - y||$.

Corollary Bellman's system of equations (Equation 4) indeed has a fixed point solution, and the iterative application of f converges to this solution.